# Detecting Out-of-Context Image-Caption Misinformation with Multimodal Embeddings

DATA5500 Senior Design Project Proposal | Fall 2026 | Instructor: Weijie Pang | Student: Mohamed Mohamed | September 2026

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## Problem Statement

A real photograph paired with a false claim about what it shows is out-of-context (OOC) misinformation, one of the hardest forms of misinformation to catch because the image itself is never altered. Existing detection tools focus on image forensics (pixel-level manipulation) or text-based fact-checking in isolation; neither approach can catch a genuine photo paired with a caption that lies about its context. This project builds a system that reasons across both modalities at once to flag exactly this kind of mismatch, a real and documented tactic in modern disinformation campaigns.

## Objective

Develop and evaluate a multimodal classifier that measures the semantic alignment between an image and its caption to detect out-of-context pairings, and analyze which categories of contextual mismatch (person, scene, or event substitution) are hardest to detect, producing both a statistically validated model and a public-facing interactive demo.

## Data & Approach

- **Dataset:** NewsCLIPpings (Luo et al., 2021), a published benchmark of real news images and captions with automatically generated out-of-context pairs, labeled by mismatch type.
- **Embedding:** Pretrained CLIP (OpenAI) produces joint image/text embeddings; cosine similarity between them forms the core detection signal.
- **Modeling:** A staged approach consisting of a similarity-threshold baseline, followed by a trained classifier (logistic regression, then a lightweight neural network) on top of the embeddings.
- **Evaluation:** Precision/recall/F1/ROC-AUC overall and broken out by mismatch type, with statistical comparison across model variants.
- **Deployment:** An interactive web application (Streamlit) where a user submits an image and caption and receives a real-time context-consistency score.

## Data Lifecycle Coverage

**Acquisition** (download & validate benchmark image/caption pairs) -> **Management** (structured storage of a curated subset) -> **Preparation/Integration** (embedding generation, label merging) -> **Analysis** (exploratory comparison of similarity distributions) -> **Modeling & Deployment** (staged classifiers, live demo app) -> **Visualization** (performance dashboards, embedding-space plots).

## Scope and Timeline

I am building this on a pretrained model rather than training one from scratch, and I am drawing on an established, publicly available benchmark rather than live web scraping. This keeps the technical scope realistic for me to complete on my own in one semester, while still giving me real data engineering, modeling, and statistical evaluation work to do. My target timeline is acquisition & baseline (weeks 1–5), modeling & design documentation (weeks 5–9), demo build & evaluation (weeks 9–13), then presentation & final documentation (weeks 13–15).